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Benchmarking LLM Robustness: Lessons from Building Plexiglass

Design decisions behind an open-source adversarial testing toolkit, and why reproducible benchmarks beat one-off red teams.

ABSTRACT

Red-teaming a language model once tells you where it stood on a Tuesday. Models change, attacks evolve weekly, and a finding that cannot be re-run is a finding that silently expires. Plexiglass is our open-source answer: a small CLI toolkit that makes adversarial testing of LLMs repeatable enough to live in a development workflow. This paper records the design decisions behind it — interactive probing versus batch scanning, live adversarial corpora versus frozen test sets, simple leakage and toxicity metrics versus composite scores — and the lessons, including the uncomfortable ones, from running it against real models. We argue that the unit of progress in LLM security is not the clever exploit but the boring, versioned, re-runnable benchmark.

1. Why reproducible benchmarks

The dominant mode of LLM security testing today is the engagement: a red team probes a model for a fixed period and delivers a report. Engagements are valuable and insufficient. They are point-in-time measurements of a moving target — the model updates, the system prompt changes, a new jailbreak family circulates, and the report describes a system that no longer exists. Meanwhile the software industry solved the analogous problem decades ago: nobody pentests a codebase once and declares it secure; we write tests and run them on every change.

Our design goal for Plexiglass was therefore not novelty of attack but repeatability of measurement: the security test as a build step. That single goal drove nearly every decision below.

2. Design decisions

2.1 A CLI first

Plexiglass ships as a command-line tool before it is a library or a platform. CLIs compose with the workflows engineers already have — shell scripts, CI runners, cron — and demand no dashboard, account, or deployment to try. The cost of a security tool is dominated by adoption friction; we optimised there. A standalone Python module and richer reporting sit on the roadmap, but the CLI is the contract.

2.2 Two modes, mapping to two questions

Testing practice kept reducing to two distinct questions, so the tool has two modes. **llm-chat** answers "what does this model do under this conversation?" — an interactive session where each response is scored against predefined metrics, currently toxicity and PII detection. It is the exploratory mode, the one a researcher reaches for when characterising a new model or reproducing a reported failure. **llm-scan** answers "how does this model perform against a corpus?" — batch benchmarking against open-source adversarial datasets, producing comparable results across models and across time. It is the regression mode, the one that belongs in CI.

2.3 Live corpora over frozen test sets

A static jailbreak test set decays in months: prompts leak into training data, and attackers move on. Plexiglass instead scrapes current adversarial prompt collections — public jailbreak repositories and datasets such as `wiki_toxic` — accepting a trade-off we consider honest: results are less comparable across long horizons, but they measure resistance to attacks people are actually using. We version corpus snapshots per run so that any result remains reproducible against the corpus that produced it.

2.4 Metrics that admit their simplicity

We score narrow, interpretable signals — toxicity of outputs, presence of PII — rather than a composite "safety score." Composite scores invite ranking; narrow metrics invite investigation. When a scan flags PII leakage, an engineer knows exactly what to look at. When a model scores 71/100 on safety, nobody knows anything. The metric set is deliberately extensible, because every deployment's definition of harm is different.

3. Lessons, including uncomfortable ones

- **The baseline is embarrassing everywhere.** The highest-value findings were rarely exotic. Models shipped behind products routinely failed against months-old public jailbreak templates — failures any scheduled scan would have caught.
- **Determinism is a myth; report distributions.** Sampling temperature, provider-side updates, and prompt formatting shift results run to run. Single-run results overclaim; we learned to run repeats and report rates, not verdicts.
- **Benchmark contamination cuts both ways.** Public adversarial prompts end up in training data, making models look robust against exactly the corpus you test with. Live corpora mitigate but do not eliminate this; genuinely novel probes remain the red team's irreplaceable contribution.
- **Scoring is a model problem too.** Toxicity and PII detectors are themselves imperfect classifiers with their own failure modes. A benchmark is only as honest as its judge, and judge disagreement belongs in the report, not under the rug.
- **Open source is a security feature.** Public methodology invites the scrutiny that proprietary evaluation escapes. Several of the tool's better decisions arrived as issues and pull requests from the SafeLlama community.

4. Where this goes

The roadmap follows the thesis. Adversarial prompt templates in llm-chat, so exploratory findings convert directly into regression tests. HTML reporting, so results travel to people who do not read terminals. A production API, so scanning attaches to deployment pipelines rather than laptops. And a broadening metric set as the community's definition of "vulnerability" matures beyond toxicity and leakage toward injection resistance and tool-misuse propensity — the frontier our other research (W4-R-2026-004) maps.

The through-line is unchanged: measurement you can re-run beats findings you can only remember. If the field converges on shared, versioned, boring benchmarks — ours or anyone's — LLM security will finally have what every other security discipline has: a way to know whether things are getting better.

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